



Dr. Keyur Patel (B.Engg, M.Tech, PhD, MIEEE)

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Professional Summary

- Eight years of experience in the field of teaching, reseach, and Industry.
- **Currently Working as Head of Research with School of Information Technology, Artificial Intelligence and Cyber Security (SITAICS), Rashtriya Raksha University, Ministry of Home Affairs, An Institute of National Importance, Government of India, Gandhinagar, Gujarat.**
- Worked with the Department of Computer Engineering, Unitedworld School of Computational Intelligence, Karnavati University, Gandhinagar, from July 2022 to Sep 2023.

- Worked with Department of Computer Science and Engineering, Institute of Advanced Research, Gandhinagar, from June-2020 to July 2022.
- **Board of Studies (BOS)** member for the curriculum design of B.Tech Computer Science specialisation in (AI, Machine Learning, Cyber Security, Software Engineering) M.Tech (IT) and BCA programmes at IAR university.
- Developed state of the art **Centre Of Excellence (COE) code unnati lab** on emerging technology (**AI, ML, IoT**) with the support of **SAP india pvt. Ltd** at Computer science department, Karnavati University.
- Worked as Main school coordinator for **ISRO-IIRS Outreach Programme for the Karnavati University**
- Worked as Main school coordinator for **Academia- Industry collaboration**.
- Recently conducted workshop on **IoT hands-on training** with the help of Industry expert.
- Delivered expert talk on **IoT and Fog Computing Enabled Smart Transportation Management System**, Student research work organised by karnavati university.
- **Experience in** Guiding Msc Data Science students, PhD research supervisor, Mentoring undergraduate students for robofest event.
- **Recently Received seed money grant for developing IoT based Project on smart parking guidance system.**

Major Role in the SITAICS, RRU, An Institute of National Importance

IoT (IoT/OT Security) and Wireless Sensor Lab Development:

- Spearheaded the development of a state-of-the-art Internet of Things (IoT) lab approx. cost around 25 Lakhs at Rashtriya Raksha University (RRU), within the School of Information Technology, Artificial Intelligence, and Cyber Security.
- Envisioned and implemented a cutting-edge facility designed to serve as a future center of excellence, driving innovation and research in IoT, AI, and IoT/OT Security.

PhD Coordinator(Head of Research):

- Served as PhD Coordinator(Head of Research) for the School of Information Technology, Artificial Intelligence, and Cyber Security (SITAICS) at Rashtriya Raksha University (RRU), overseeing the academic and administrative aspects of the PhD program.
- Facilitated effective communication between doctoral candidates, faculty, and administration, ensuring smooth progress and resolution of academic issues.
- Organized RAC for all aspects of PhD candidates.
- Assisted in the development and implementation of program policies, ensuring compliance with university standards and academic regulations.

- Provided guidance and support to PhD students throughout their research journey, from admission to thesis submission and defense.

Coordinator AI Tech Shaksham Program for all pre-final Year Students:

- Successfully implemented the AI Tech Shaksham Program in collaboration with Edunet and Microsoft, benefiting all pre-final year students of the School of Information Technology, Artificial Intelligence, and Cyber Security (SITAICS) at Rashtriya Raksha University (RRU).
- Facilitated Internship and Placement assistance for students, successfully securing opportunities in renowned companies.

Summary of other roles at different level in SITAICS, RRU

- B.Tech Program Coordinator
- Main faculty Member for R& D and Consultancy
- Faculty Member Admission Committee
- Faculty Member Industrial Visit
- Faculty IQAC Committee
- Faculty Member Internship & Placement Committee
- **Course Coordinator of B.Tech Electronics(VLSI Design and Technology) Program in the School of Applied Sciences, Engineering and Technology**

Role at University Level

- Currently working as a **Member** of University **Doctroal Research Committee**

Delivery of Expert Talk(Resource Person)

- Invited as a Resource Person : National Commission for Women (NCW), Government of India sponsored by Capacity Building Programme at the School of Criminal law and Military Law (SCLML), RRU, Gandhinagar: **Title of the topic: Tech Horizons: Navigating Emerging Opportunities in AI, Drone Technology.**
- Delivered Expert talk on the topic of **Basic Networking essentials for Cyber Security** to Nepal Police at Rashtriya Raksha University, Gandhinagar.
- Invited as Resource person of the FDP titled **CyberSecML : Exploring the Fusion of Security and Machine Learning, Jointly Organized by RRU and IIIT Kottayam.**

- Spearheaded the successful development and establishment of a state-of-the-art Center of Excellence Lab on emerging technology, securing a substantial 20 lakhs grant from SAP India Pvt Ltd for the Unitedworld School of Computational Intelligence(USCI), Karnavati University, Gandhinagar
- Led the strategic planning and execution of the Center of Excellence, facilitating advanced research and innovation in cutting-edge technologies, supported by the generous funding from SAP India Pvt Ltd.
- Collaborated with key stakeholders, including SAP India Pvt Ltd and university administrators, to secure the significant 20 lakhs grant, elevating USCI as a premier hub for emerging technology research.
- Successfully established valuable partnerships with industry experts and technology leaders, enabling knowledge exchange and real-world applications of research findings.
- Utilized the 20 lakhs grant effectively to equip the Center of Excellence with cutting-edge equipment and resources, empowering researchers and students to explore innovative solutions in emerging technology domains.
- The Center of Excellence's establishment and research achievements significantly elevated the university's reputation, attracting further funding opportunities and collaborative endeavors with leading industry players.
- **Successfully designed course curriculum focused on emerging technology under the roof of this COE lab for undergraduate students with the help of SAP officials.**
- By leveraging the generous support from SAP India Pvt Ltd, the Center of Excellence Lab has become a key driver in fostering technological advancement, producing skilled graduates, and making notable contributions to the emerging technology landscape.

Technical Skills

- Computer Skills & Programming Languages:- **C, Matlab and Simulink, Embedded C, Arduio (ATMega series- C Programming), ARM CORTEX-M series (C-Programming), Labview, Python, VHDL, Ladder Logic, LATEX.**
- Assembly Languages: - 8085(Microprocessor), Microcontroller(8bit/16bit/32bit), ARM7, ARM cortex-M series.
- Software tools:- Xilinx ISE 9.2, Quartus-II, keil Vision4, Spice(TI), **pSpice (Analog circuits design), Proteus professional(Virtual Hardware design), Google colab, Jupyter notebook(Python Programming).**
- Operating Systems: - Windows 98, 2000, XP, Vista, Windows 8, Windows 10, Linux.

Research Interest

- Robotics and AI
- Internet of Things (IoT)

- Industrial Internet of Things (IIoT)
- Cyber Physical System
- Multi-agent system
- Artificial Intelligence for Robotics Application
- Machine Learning
- Deep Learning
- Aerial Computing
- Embedded Computing
- Signal Processing
- Computer Vision
- Machine Vision
- Drone Forensic
- High Performance Computing (HPC)
- Autonomous system e.g UAV, Quad-Copter Robot

Research major funded projects

- **RFID authentication-based smart parking guidance using IoT: An implementation point of view -(KU/2022/SMS/004)**
 Funded by : Karnavati University, Gandhinagar
 Budget(Seed money grant) : 1,01,001/-
 Role : PI
 Tenure : 1 years(2022-2023)
- **IoT and Fog computing enabled advance public bus transport management system(APBTMS): An Innovative smart bus concept**
 Funded by : Karnavati University, Gandhinagar
 Budget(Seed money grant) : 1,54,975/-
 Role : PI
 Tenure : 2 years(2023-2024)
- **IoT and Machine Learning enabled automated soil nutrient monitoring for smart agriculture: A smart techno-agri concept-(KU/2022/SMS/009)**
 Funded by : Karnavati University, Gandhinagar
 Budget(Seed money grant) : 1,01,001/-
 Role : Co-PI
 Tenure : 2 years(2023-2024)

Recent Research Project Going On

- **AI-enabled automated detection of Alzheimer's disease**
- **AI-assisted Leader-following consensus of multiple heterogeneous robotic systems: A real-time multiapplication perspective**
- **Visual Inspection in Manufacturing Industry Using AI**
- **IoT-based Electric Kick Scooter**
- **Voice-assisted Smart Automation System using AI**
- **IoT enabled smart contactless door**
- **IoT-based Smart Inverter**

PG Guided Project Detail

- Guided PG student on the project titled “**Classification of the Sismic Events through Machine Learning**” jointly with Institute of Seismological Research, Department of Science and Technology, Gandhinagar

Recent Work Experience

Assistant Professor (22th Sep 2023- Till Date):

- Working with **School of Information Technology, Artificial Intelligence and Cyber Security (SITAICS), Rashtriya Raksha University(RRU), Ministry of Home Affairs, An Institute of National Importance**, Government of India, Gandhinagar, Gujarat
- Delivered lectures on the subjects **Machine Learning**, and **Speech Information Processing**, for postgraduate M.Tech Data Science and Machine Learning students.
- Delivered lectures on the subjects **Digital Systems**, **Internet of Things**, and **Deep Learning**.
- Delivered lectures on **Basic Networking essentials for Cyber Security** to Nepal Police at Rashtriya Raksha University, Gandhinagar.

Assistant Professor (20 July 2022 - 20th Sep 2023):(1 Year 2 months)

- Worked as an **Assistant Professor** in the department of **Computer Science** at **United-world School of Computational Intelligence, Karnavati University**, Gandhinagar.
- Delivered lectures on the subjects **Computer Networks**, **Industry Elective (Collaborative)** and **Human Computer Interaction (HCI)**, **Electrical and Electronics** for undergraduate computer science students.
- Department main faculty **Industry collaborative coordinator** for academia- Industry Interaction for the student technical visit and Internship.

- Department main faculty coordinator for the **Code Unnati** center of excellence on emerging technologies e.g AI, ML, IoT.
- **Assistant Professor (1 June 2020 - 19 July 2022):(2 Years 2 months)**
- Worked as an **Assistant Professor** and an **Academic coordinator** of B.Tech **Computer Science, Information Technology, and Information and Communication Technology** in the department of **Computer Science and Engineering** at the Institute of Advanced Research(IAR), Gandhinagar.
- Worked as a Program Leader of B.Tech **Information and Communication Technology (ICT)** in the Department of Computer Sceince and Engineering at IAR.
- Worked as a **Lab-incharge** for the development of **IoT(Product and Design)** Lab at IAR.
- Recently developed Curriculum for B.Tech ICT at IAR.
- **Board of Studies (BOS)** member of B.Tech Computer Science specialisation in (AI, Machine Learning, Cyber Security, Software Engineering) M.Tech (IT) and BCA programmes at IAR university.
- Delivered lectures on the subjects **Microprocessor and Microcontroller, Computer Network,Computer Architecture, Digital Logic Design, Software Engineering, Signal and System, Embedded System, and Basic Electronics** for the undergraduate engineering students.
- Delivered lectures on **Signal Processing** course to Postgraduate (Data Science) students.
- Delivered lectures on the subject **Mathematical tools for computation** for the PhD students of IAR.

Latest Research Achievement

Conference Paper

- **Keyurkumar Patel, D. Patel, N. Chaudhary, D. Thakkar, P. Savaliya, A. Mistry, Advance Public Bus Transport Management System: An Innovative Smart Bus Concept** accepted in **IEEE 42nd International Conference on Consumer Electronics, Las Vegas, NV, USA.**
- **R. Maliya, Dr Keyur Patel, An Overview of Safety Using Information Routing Protocol in Vehicular Ad Hoc Network,** Accepted in **IEEE R10 HTC 2023.**

Book Chapter

- **Keyurkumar Patel, P.Sunnapu, S. Gautam, Emerging Technologies in Fintech : A Case Study,** Robotics and Automation in Industry 4.0, Bentham Science, Accepted.

- **Keyurkumar Patel**, M. Kandasamy, R. Shanmugam, T. Uttare, N. Samani, M. Aqeel, S. Joshi, **RFID authentication-based Intelligent Parking Management System using IoT and Mobile Applications: An Implementation point of view**, Green Computing, CRC press, Taylor and Francis.
- M. Kandasamy, R. Shanmugam, S. Saha, R. Patil, **Keyur Patel**, Interfacing of Actuators and Sensors, Cyber-Physical Systems and Supporting Technologies for Industrial Automation, IGI Global, DOI:10.4018/978-1-6684-9267-3.

Intellectual Property

- K.Anbazhagan, S. Rathod, **Keyurkumar Patel**, et al. **Smart Inhaler with Real-time Inhalation pattern Analysis and Method**, Indian Utility Patent, Publish Date: 29 December 2023.
- P. Selladurai, P. Bilgaiyan, R. Vidhyavathi, J. Sahayarayan, **Keyurkumar Patel**, **Smart pulse oximeter with data storage and wireless access**, Design number - 6318072, Grant Date: 17 October 2023.
- A. Gupta, R. Shanmugam, M. Kandasamy, R. Alam, D. Mohan, **Dr Keyurkumar Patel**, **A Speech Coding Method For Speech Recognition Employing Convolution Neural Networks**, Indian Patent, Application Number-202221054497, Filing Date -23/09/2022-Indian Patent Office.
- Dr V. Singh, Deepthi K, **Dr Keyurkumar Patel**, Dr N Sankaran, F. Ahmad, Dr P Shelke et al., **Smart Vein Identifying Wrist Band**,UK design patent, Design Number - 6303703, Grant Date : 24 August 2023.
- Dr.H Kumar, **Dr keyurkumar Patel** et al., **Fish Disease Prediction Device in Aquaculture**,UK design patent, Design Number -6304705, Grant Date 01 September 2023.
- M. Kandasamy, D Bhavani, A. Gupta, **Dr Keyurkumar Patel**,**Design of IoT based Automatic Star Climbing Material Handler Robot**, Canadian Copyright, Reg. Number : 1196588
- Dr A. Oza, Dr N. Das, **Dr Keyurkumar Patel**, C. Chaudhari, Dr K. Bhandari, S. Ramkumar, Dr A. Kumar, **IoT Health ATM** Industrial Design Patent under review.

Online Certification Courses and latest FDP

- Professional Certification in **AI for Medical Diagnosis** authorized by Coursera.
- Successfully completed workshop on **AI application on VLSI Design and Technology** organised by DAIICT, Gandhinagar, Gujarat.
- Successfully completed FDP on **Implementing IoT & Machine Learning application on Raspberry Pi** organised by Department of E&TC BSCER, Pune in association with industrial partner Smart Logic Technologies, Pune.

- Successfully completed winter school on **Cybersecurity and Blockchain 2023** organised by IEEE Computer Society, Gujarat.
- Completed five days online STTP on **AI based signal and Image Processing** organized by NIT, Jalandhar, Punjab.
- Completed five days online FDP on **Cyber Security (Emerging Threats and Trends against Digital Defenses)**, ICFAI Tech University, Jaipur.
- Attended Short term program on **Research-o-Pedia 3.0: Writing to Publication** organised by Karnavati University, Gandhinagar.
- Successfully attended FDP on **National Education Policy (NEP) 2020: Implementation and Transformation** organised by Karnavati University, Gandhinagar.
- Successfully completed online training course titled “**Machine Learning**” on Internshala online platform.
- Successfully completed online training courses titled “**Embedded Systems Programming on ARM Cortex - M3/M4 Processor**” on Udemey online platform.

Major Project and Experiments

- Design and mathematical modeling for quadcopter (Unmanned aerial vehicle).

Webpage Link

- <https://scholar.google.com/citations?hl=en&user=7FIONVoAAAA>
- https://www.researchgate.net/profile/Keyur_Patel41

GATE -2015 (Qualified), Stream:-Electronics and Comm.Engg

Gate marks out of 100	27.19
Gate score	377

Academic Qualification

2016 - 2020	Doctor of Philosophy (PhD) From the Department of Electrical and Computer Science, Institute of Infrastructure Technology Research and Management (IITRAM) (Established by Gov.of Gujarat) and Mentored by IIT-Gandhinagar , Ahmedabad (Course work CPI 8.83)
2012 - 2014	Master of Technology in Control and Automation(Drone Technology) , (CPI: 7.9/10 (First class with distinction), Department of Electronics Engg , Institute of Technology, Nirma University, Ahmedabad
2005 - 2009	Bachelor of Engineering in Electronics ; 64.32%(First class), BVM gov. engineering college , Vallabh Vidhyanagar, Gujarat
2004 - 2005	Higher Secondary in Science (GSHEB); 83.11%, Kisan Madhaymik Vidhyalaya, Lunawada, Gujarat
2003 -2004	Secondary school (GSHEB); 82.71%, Omkar Maulik Vidhayalaya, Lunawada, Gujarat

PhD thesis title : Discrete-time Sliding Mode Protocols for Leader-Following Consensus of Discrete-time Multi-Agent System.

List of Publications in Peer-Reviewed Journals and Reputed Conferences

- **Keyurkumar Patel** and A. Mehta, *Discrete-time Sliding Mode Protocols for Leader-following consensus of Discrete Multi-Agent System with Switching Graph Topology* **European Journal of Control**, Volume 51, 2020, Pages 65-75, ISSN 0947-3580, <https://doi.org/10.1016/j.ejcon.2019.06.011>.
- **Keyurkumar Patel** and A. Mehta, *Event triggered Consensus of Homogeneous 2-DOF Robotic Arms using Discrete Higher Order Super-twisting Algorithm*, **European Journal of Control**, Volume 56, 2020, Pages 231-241, ISSN 0947-3580, <https://doi.org/10.1016/j.ejcon.2020.01.006>.
- Dipesh Shah, A. Mehta, **Keyurkumar Patel**, Andrzej Bartoszewicz, *Event-Triggered Discrete Higher-Order SMC for Networked Control System having Network Irregularities* in **IEEE Transactions on Industrial Informatics**, vol. 16, no. 11, pages. 6837-6847, Nov. 2020, doi: 10.1109/TII.2020.2973739.
- **Keyurkumar Patel** and A. Mehta, *Fixed time steps discrete-time sliding mode consensus protocols for two degree of freedom helicopter systems*, **Transactions of the Institute of Measurement and Control**, vol. 43, no. 15, pages. 3322-3334, April 2021. doi:10.1177/01423312211004778.

- **Keyurkumar Patel** and A. Mehta , *Discrete-time Sliding Mode Control for Leader Following Discrete-time Multi-Agent System*, **IECON 2018** - 44th Annual Conference of the **IEEE Industrial Electronics Society**, Washington, DC, 2018, pp. 2288-2292.
- **Keyurkumar Patel** and A. Mehta *Discrete Higher Order Sliding Mode Protocol for Leader-Following Consensus of Heterogeneous Discrete Multi-Agent System*, In: Mehta A., Rawat A., Chauhan P. (eds) *Advances in Control Systems and its Infrastructure. Lecture Notes in Electrical Engineering*, vol 604. Springer, Singapore.
- Jatin savliya, **Keyurkumar Patel**, A. Mehta, *Distributed event-triggered sliding mode control for voltage synchronization of DC microgrid using leader-follower consensus protocol*, In: Mehta A., Rawat A., Chauhan P. (eds) *Advances in Control Systems and its Infrastructure. Lecture Notes in Electrical Engineering*, vol 604. Springer, Singapore.
- **Keyur Patel** and A. Mehta, *Second Order Sliding Mode Control of Active Suspension System with Super-Twisting Algorithm and Disturbance Observer*, 43th annual conference of the **IEEE Industrial Electronics Society**, pp.6526-6531, October-2017, Beijing-China.
- Sandeep Sharma, Dhananjay Srivastava, A. Mehta and **Keyur Patel**, *Design and Implementation of Second Order Sliding Mode Controller for 2-DOF Flexible Robotic Link*, 2nd **International Conference on Power, Energy and Environment: Towards Smart Technology (ICEPE)**, NIT-Meghalaya, Shillong, India, 2018, pp. 1-6.
- Dhananjay Srivastava, Sandeep Sharma, **Keyur Patel** and A. Mehta, *Second Order SMC Controller Design for Rotary Inverted Pendulum: An Under Actuated System*, 2018 2nd **International Conference on Power, Energy and Environment: Towards Smart Technology (ICEPE)**, Shillong, India, 2018, pp. 1-6.
- **Keyur Patel** and Jayesh Barve, *Modeling, Simulation and Control Study for Quad-Copter UAV*, **IEEE International conference on industrial and information system**, vol.9, pp.1-6, Dec-2014 , IIITM-Gwalior.
- Jayesh Barve, **Keyur Patel**, *Modelling, Simulation and Altitude-Range-Analysis of Quad-copter UAV*, **IFAC Proceedings Volumes**, Volume 47, Issue 1, 2014, Pages 1126-1130, ISSN 1474-6670, ISBN 9783902823601, <https://doi.org/10.3182/20140313-3-IN-3024.00209>.

Book

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- **Keyurkumar Patel** and **A. Mehta**, *Discrete-Time Sliding Mode Protocols for Discrete Multi-Agent System*, Springer Singapore, ISBN - 978-981-15-6310-2, August-2020.

Book Chapter

- **Keyurkumar Patel** and **A. Mehta**, *Discrete Higher-order Sliding Mode Leader-Following Consensus Protocols for Homogeneous Discrete-time Multi-Agent System*, accepted in Emerging trends on sliding mode control.

Journal Reviewer

- ISA Transactions
- Biomedical Signal Processing & Control
- IEEE Transactions on Industrial Informatics
- IEEE Transactions on Industrial Electronics

Other Work Experience - Academic and Industrial (More than 4 years)

Assistant Professor (16 June 2014 - 30 July 2016):

- Worked in the Electronics and Electrical Engineering Department, SVBIT engineering college Gandhinagar-Gujarat(2 Year 1month 14 days).
- Delivered lectures and conducted labs on the subjects control system engineering, microprocessor and microcontroller, and signal and system for the undergraduate electronics and electrical engineering students.
- Worked as class-coordinator of 2nd year electronics and electrical engg students to take care about their personal to professional carrier including meeting with their parents for their better future.
- Worked as design engineering subject coordinator for undergraduate electronics and electrical engineering students newly introduce course by GTU.
- Project guide titled “Mathematical Modeling, Simulation and Control of DC motor using Matlab simulink” to two final year undergraduate students.

Project Engineer- Railway Signalling (22 Oct. 2009 to 1 Nov. 2011):

- Worked with INSTRUMENTATION LTD. - KOTA (A Govt. Of India Enterprise), at its BARODA office as a Project Engineer for more than 2 years.
- Worked as the post of a Project Engineer in Railway Department under the project of “Railway Signal Automation” between Godhra-Baroda Railway Line.
- Successfully Designed, Installed, Tested and Commissioned the SSI (Solid state interlocking) System at Chapaner, Bakrol, pilol, Kharsaliya stations in Western Railway.
- Designed - logic and physical ckt. for vizag steel plant-B.F.yard at Vishakhapatnam. It is the SSI system working on K5-B (Kyosan – A Japan based Company) Technology.
- Sound knowledge in Design/Checking and Verification of Signalling Plans, Interlocking circuit diagrams, location drawings, Interlocking Charts and Logic expression.
- Successfully done the implementation and designing of IBS (Intermediate block signal) related ckt. in SSI system, testing of cables. Sound knowledge of various aspect of signal, point operation at field site.

- Verification & debugging of logic program as per RDSO(Research design & standard organization) rules given by the ministry of railway.

Workshop and Conference

- Participated three days short term training programme on **High Performance Computing and Deep Learning** held at DAIICT, Gandhinagar during 26 – 28th May 2022.
- Participated and contributed as main **student coordinator** in short term Training Program on “**Design and Implementation of Advanced Controllers**” held at IITRAM during 21st- 25th January, 2019.
- Participated and contributed as **student coordinator** in short term Training Program on “**Advanced Control Techniques and Their Applications**” held at IITRAM during 27th Nov. - 1st Dec., 2017.
- Participated in the course on “**Advanced sliding mode control and estimation for real complex systems of the 21st century**” held at IIT-Roorkee during 24st-27th October, 2017.
- Participated in the short course on “**Data Analytics: Applications in Modelling, Control and Signal Processing**” held at IIT Gandhinagar on 25st-26th October, 2013.
- Attended the 3rd Nirma University International Conference on Engineering, NUiCONE-2012 organized by Institute of Technology, Nirma University, Ahmedabad in association with IEEE and science direct.

Seminar and FDP

- Participated a Design Engineering faculty development programme at GTU, L.D college of engineering-Ahmedabad during 4 august 2015 to 7 august-2015.
- Participated a National Seminar on “Wireless Technology in Automation: Current and Future Trends” on 20th April,2013 conducted by Instrumentation and Control Engineering branch, Electrical Engineering Department, Institute of Technology, Nirma University, Ahmedabad.

Short Term Training of Industrial Automation

SHORT TERM TRAINING OF INDUSTRIAL AUTOMATION AND ROBOTICS BY MSME(A GOV. OF INDIA):

- Electrical hardware and process control.
- PLC programming like(siemens,ABB,Delta).
- HMI and SCADA(ABB) programming.
- Variable Frequency Drives(VFD).

- PID Controller and MOTOR Basics.
- Communication protocols.
- Control panel designing.
- Testing, installation and commissioning.

M. TECH Project Work - Drone Technology (Thesis)

Mathematical Modeling, Simulation and Control of quad copter UAV(Unmanned aerial vehicle)

Advisor: Dr. Jayesh J Barve (R&D Principal Engineer, Controls & Optimization, Electrical Technologies GE Global Research, JF Welch Technology Center, GE India Industrial Pvt Ltd., Bangalore)

Training (V.B Tech automation)

- I had been undergone training of lab view basic like OPC function, data socket troubleshooting, mod bus function.
- The field work assistance gave practical exposure to the field devices and controlling techniques and understands the process of particular control system.
- Practically learned how to configure variable in a network and transmit through networks. OPC server assignment learned how to convert communication protocol into OPC Protocol. Modbus assignment gave some brief idea about MODBUS protocol.

Graduation Level Project

- Microcontroller based Manual Landrover robot triggering and relay circuitry.

Achievements

Professional Membership

- Professional Member, Institute of Electrical and Electronics Engineers(IEEE)

Academic Achivements

- Received Certificate of Honour for presenting paper in ACODS International conference at IIT Kanpur during the academic year 2013-14 by Nirma University.
- Received first rank in M.Tech entrance exam taken by Nirma University.
- Received first rank in 10th standard at school level and also rank in science quiz at dist. level.

Extra-Curricular Activities

- Participated in “semi-qui-ro” event organized by i-maze '08 at GCET engineering college and got second rank in the event.
- Participated in “Racer’s Den” robotics event organized by Advent'07 at B.V.M. engineering college.
- Participated in “circuitrix” event organized by Advent'07 at B.V.M. engineering college.
- Sub co-coordinator of youth festival -2015 organised by Gujarat Technological University.

References

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